Infinite geometric series

1 The absolute value of the common ratio must be less than 1.

2

a i $r = 2$ ii Divergent

b i $r = -4$ ii Divergent

c i $r = \frac{1}{2}$ ii Convergent

d i $r = \frac{1}{4}$ ii Convergent

3

a i $r = \frac{1}{4}$ ii $\frac{8}{3}$

b i $r = \frac{1}{5}$ ii $\frac{625}{4}$

c i $r = -\frac{1}{2}$ ii $\frac{32}{3}$

d i $r = -\frac{1}{4}$ ii $\frac{112}{5}$

4

a $r = \frac{\sqrt{5}}{5}$

b $\frac{(25 + 5\sqrt{5})}{4}$

5

a $n = 11$

b 9

6 $\frac{1}{3}$

7

a $r = \frac{1}{2}$

b $2, \frac{1}{2}, \frac{1}{4}$

8 $\frac{2}{9}$

9

a 4

b 4

c $\frac{1}{12}$

d $-\frac{7}{4}$

e 9

f $\frac{1}{8}$

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Recurring decimals

10

a **i** $0.6 + 0.06 + 0.006 + 0.0006 + \ldots$ **ii** $\frac{2}{3}$

b **i** $0.06 + 0.006 + 0.0006 + 0.00006 + \ldots$ **ii** $\frac{1}{15}$

c **i** $0.15 + 0.0015 + 0.000015 + \ldots$ **ii** $\frac{5}{33}$

d **i** $0.021 + 0.00021 + 0.0000021 + \ldots$ **ii** $\frac{7}{330}$

11 $\frac{19}{33}$

12 **a** $0.252525\ldots$ **b** $\frac{25}{99}$

13 $\frac{994}{999}$

Algebraic series

14 $576, 144, 36$

15 $x = \frac{1}{5}$

16 $x < 8$ and $x > 6$

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